

Simple Linear Regression and Correlation

From Walpole (chap # 11)

Simple Linear Regression

- The dependence of one variable over the other variable is called “Regression”.
- The statistical method which helps us to estimate the value of dependent variable from the known value of independent variable is called regression.
- The statistical model for simple linear regression is given below. The response Y is related to the independent variable x through the equation (True regression line):

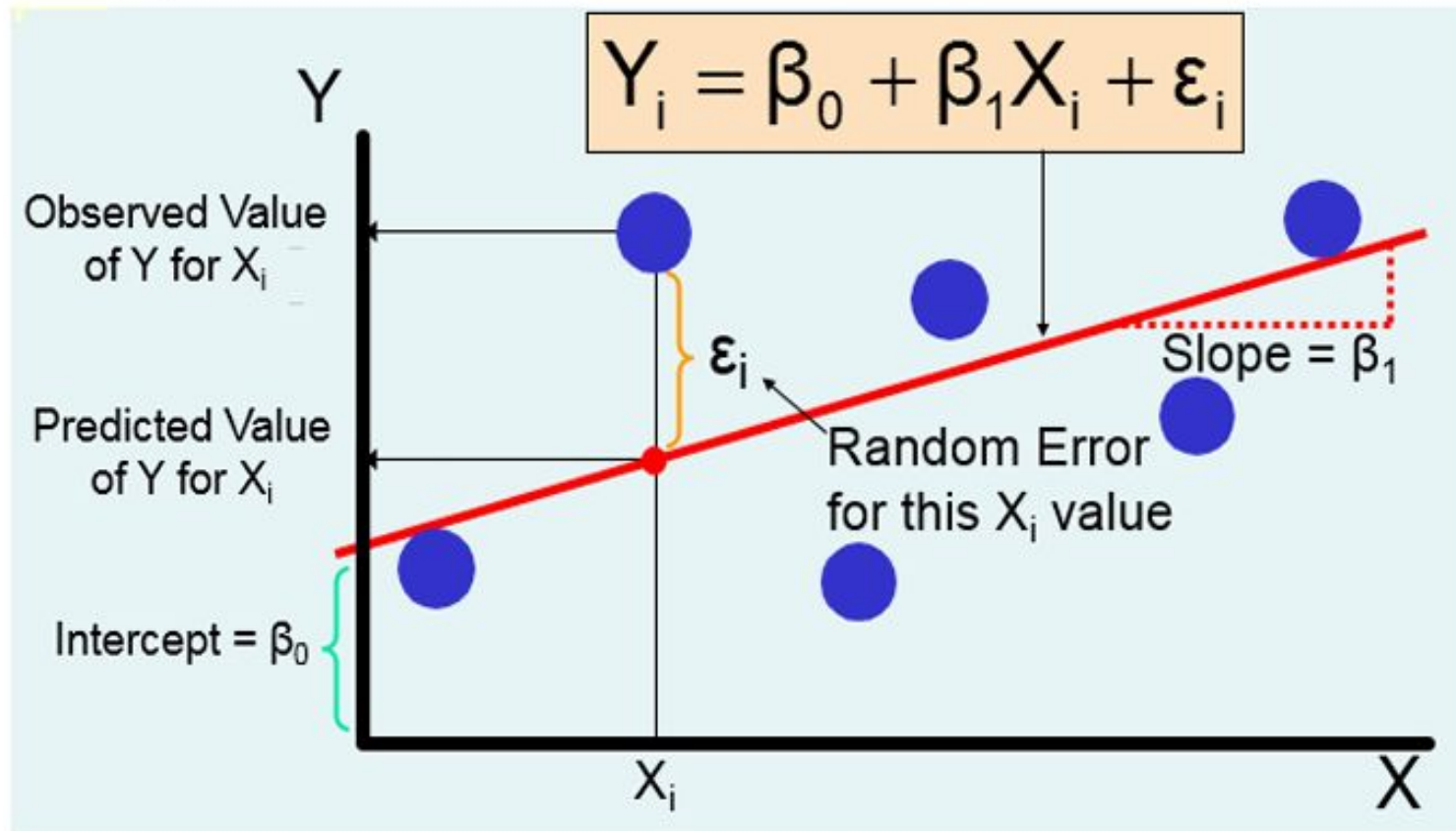
$$Y = \beta_0 + \beta_1 x + \epsilon.$$

Where:

- Y: Dependent variable
- X: Independent variable
- β_0 : Intercept
- β_1 : Slope of the line
- ϵ : Error term
- Health: Predicting blood pressure based on hours of exercise per week.
- Real Estate: Predicting house prices based on size (sq ft).
- Education: Estimating exam scores based on hours studied.

We must keep in mind that:

- In practice β_0 and β_1 are not known and must be estimated from data.
- we never observe the actual ϵ values in practice and thus we can never draw the true regression line
- We can only draw an estimated line.



Simple Linear Regression Equation as Estimator Of Population Regression Line:

The simple linear regression equation provides an **estimate** of the population regression line

Estimated
(or predicted)
Y value for
observation i

Estimate of
the regression
intercept

Estimate of the
regression slope

Value of X for
observation i

$$\hat{Y}_i = b_0 + b_1 X_i$$

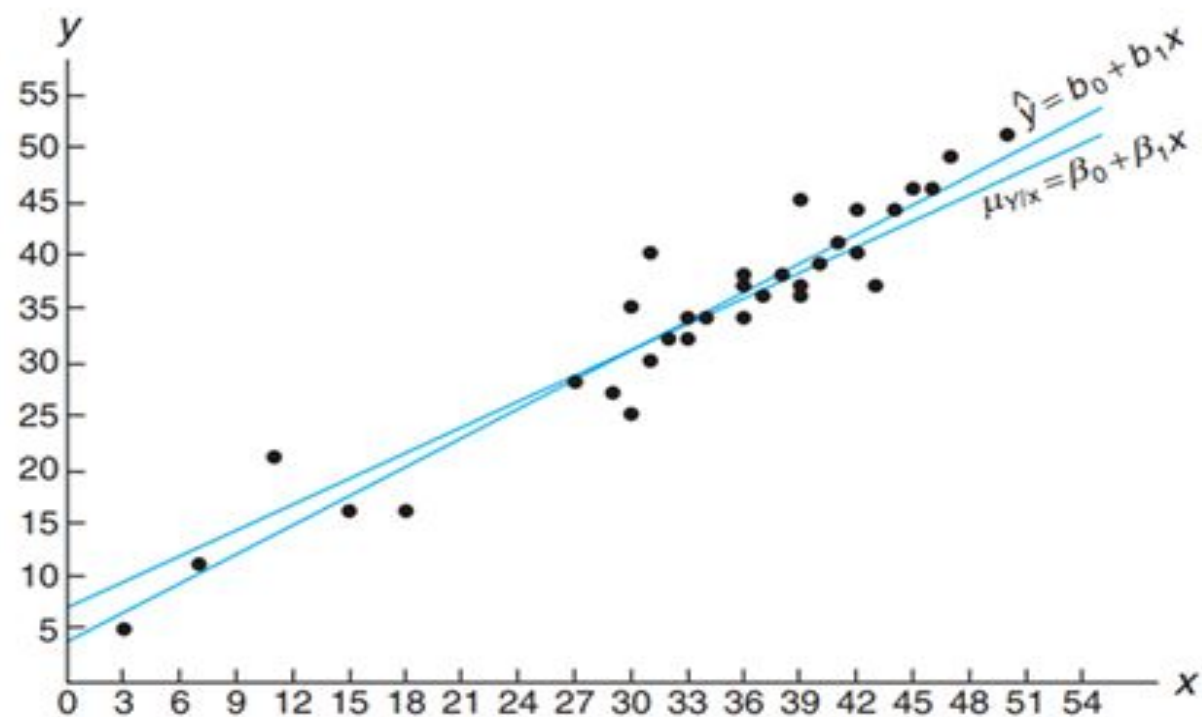


Figure 11.3: Scatter diagram with regression lines.

The fitted regression line and a *hypothetical true regression line* are shown on the scatter diagram of Figure 11.3. This example will be revisited as we move on to the method of estimation, discussed in Section 11.3.

11.3 Least Squares and the Fitted Model

Residual: **Error in Fit** Given a set of regression data $\{(x_i, y_i); i = 1, 2, \dots, n\}$ and a fitted model, $\hat{y}_i = b_0 + b_1 x_i$, the i th residual e_i is given by

$$e_i = y_i - \hat{y}_i, \quad i = 1, 2, \dots, n.$$

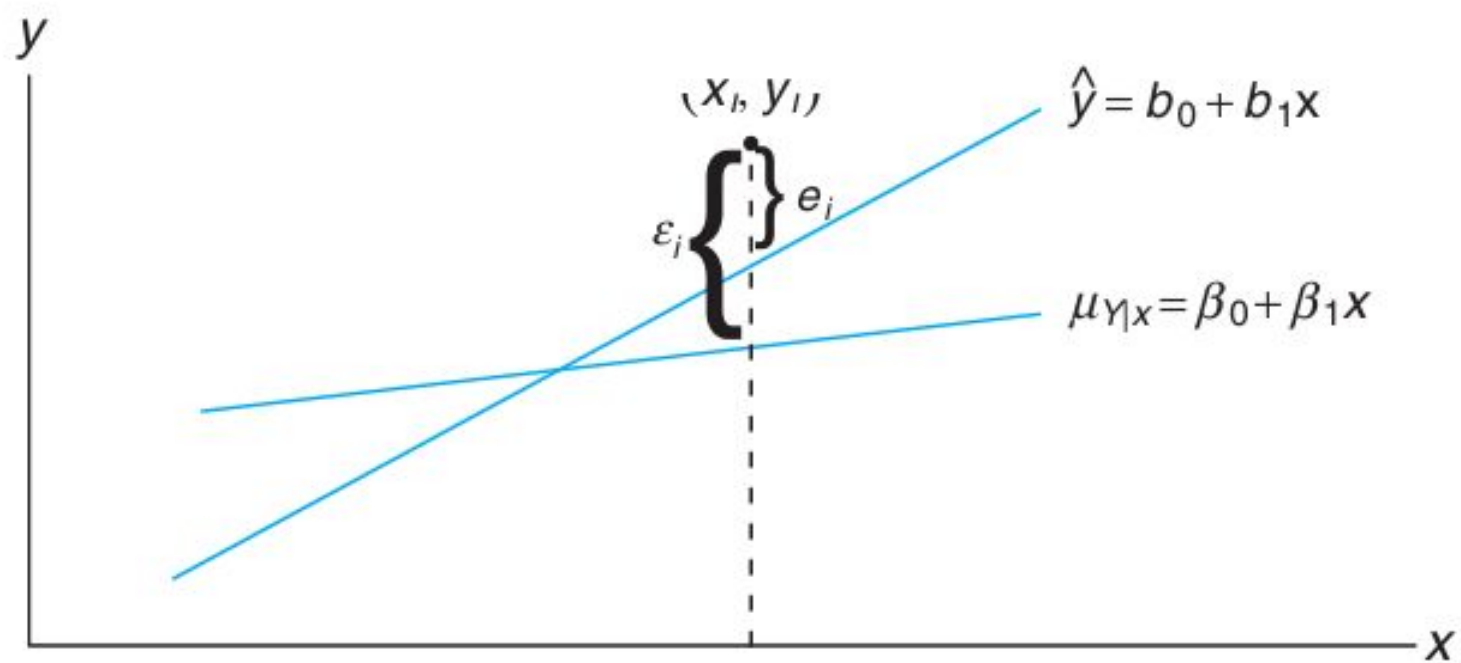


Figure 11.5: Comparing ε_i with the residual, e_i .

Least-Squares Criterion

The **least-squares criterion** is that the line that best fits a set of data points is the one having the smallest possible sum of squared errors.

We shall find b_0 and b_1 , the estimates of β_0 and β_1 , so that the sum of the squares of the residuals is a minimum. The residual sum of squares is often called the sum of squares of the errors about the regression line and is denoted by **SSE**.

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

Estimating the Regression Coefficients

Given the sample $\{(x_i, y_i); i = 1, 2, \dots, n\}$, the least squares estimates b_0 and b_1 of the regression coefficients β_0 and β_1 are computed from the formulas

$$b_1 = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \text{ and}$$
$$b_0 = \frac{\sum_{i=1}^n y_i - b_1 \sum_{i=1}^n x_i}{n} = \bar{y} - b_1 \bar{x}.$$

Quantity	Defining formula	Computing formula
S_{xx}	$\Sigma(x_i - \bar{x})^2$	$\Sigma x_i^2 - (\Sigma x_i)^2/n$
S_{xy}	$\Sigma(x_i - \bar{x})(y_i - \bar{y})$	$\Sigma x_i y_i - (\Sigma x_i)(\Sigma y_i)/n$
S_{yy}	$\Sigma(y_i - \bar{y})^2$	$\Sigma y_i^2 - (\Sigma y_i)^2/n$

Regression Equation

The regression equation for a set of n data points is $\hat{y} = b_0 + b_1 x$, where

$$b_1 = \frac{S_{xy}}{S_{xx}} \text{ and } b_0 = \frac{1}{n}(\Sigma y_i - b_1 \Sigma x_i) = \bar{y} - b_1 \bar{x}.$$

TABLE 14.3
Four data points

x	y
1	1
1	2
2	2
4	6

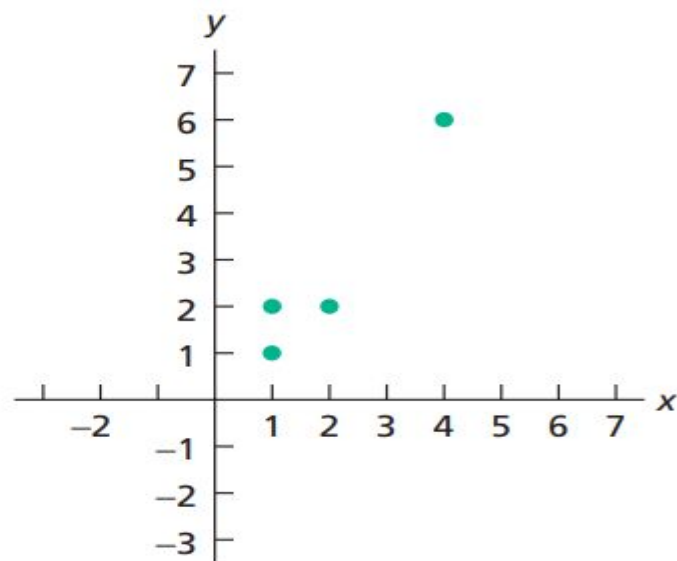
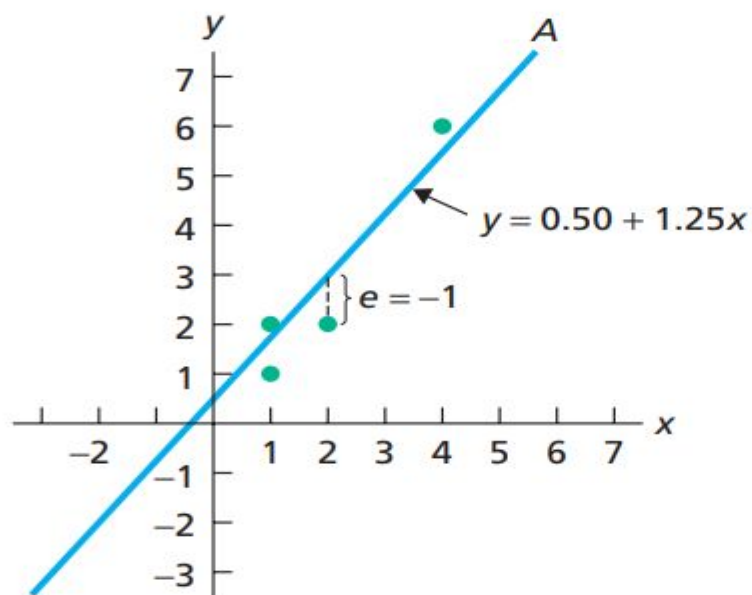


FIGURE 14.9

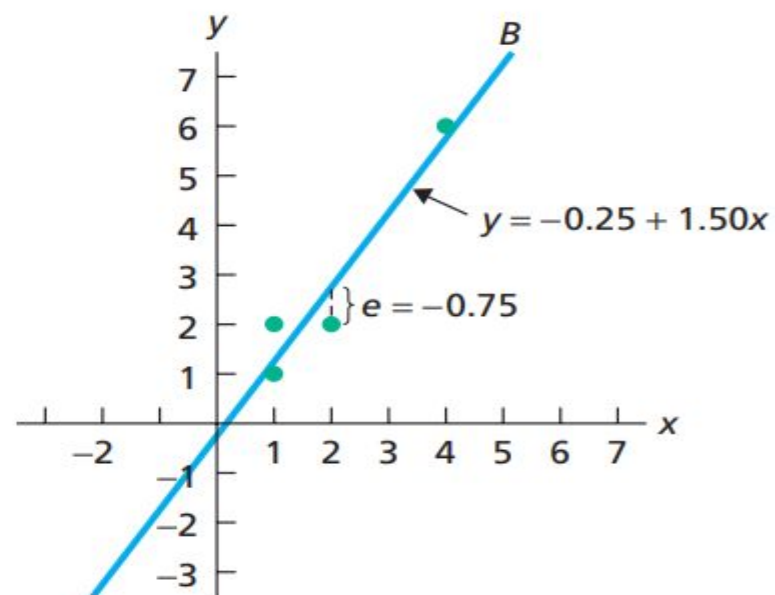
Two possible lines to fit the data points in Table 14.3

Line A: $y = 0.50 + 1.25x$



(a)

Line B: $y = -0.25 + 1.50x$



(b)

TABLE 14.4

Determining how well the data points in Table 14.3 are fit by (a) Line A and (b) Line B

Line A: $y = 0.50 + 1.25x$					Line B: $y = -0.25 + 1.50x$				
x	y	$\hat{y}$	e	e^2	x	y	$\hat{y}$	e	e^2
1	1	1.75	-0.75	0.5625	1	1	1.25	-0.25	0.0625
1	2	1.75	0.25	0.0625	1	2	1.25	0.75	0.5625
2	2	3.00	-1.00	1.0000	2	2	2.75	-0.75	0.5625
4	6	5.50	0.50	0.2500	4	6	5.75	0.25	0.0625
				1.8750					1.2500
(a)					(b)				

To decide which line, Line A or Line B, fits the data better, we first compute the sum of the squared errors, $\sum e_i^2$, in the final column of Table 14.4(a) and Table 14.4(b). The line having the smaller sum of squared errors, in this case Line B, is the one that fits the data better. Among all lines, the **least-squares criterion** is that the line having the smallest sum of squared errors is the one that fits the data best.



11.6 In a certain type of metal test specimen, the normal stress on a specimen is known to be functionally related to the shear resistance. The following is a set of coded experimental data on the two variables:

Normal Stress, x	Shear Resistance, y
26.8	26.5
25.4	27.3
28.9	24.2
23.6	27.1
27.7	23.6
23.9	25.9
24.7	26.3
28.1	22.5
26.9	21.7
27.4	21.4
22.6	25.8
25.6	24.9

- (a) Estimate the regression line $\mu_{Y|x} = \beta_0 + \beta_1 x$.
- (b) Estimate the shear resistance for a normal stress of 24.5.

	x	y	xy	x^2
	26.8	26.5	710.2	718.24
	25.4	27.3	693.42	645.16
	28.9	24.2	699.38	835.21
	23.6	27.1	639.56	556.96
	27.7	23.6	653.72	767.29
	23.9	25.9	619.01	571.21
	24.7	26.3	649.61	610.09
	28.1	22.5	632.25	789.61
	26.9	21.7	583.73	723.61
	27.4	21.4	586.36	750.76
	22.6	25.8	583.08	510.76
	25.6	24.9	637.44	655.36
Total	311.6	297.2	7687.76	8134.26

$$b_1 = \frac{(12(7687) - (311.6)(297.2))}{(12(8134.26) - 311.6^2)} = -0.6860$$

$$b_0 = \frac{297.2}{12} - \frac{(-0.6860)(311.6)}{12} = 42.5818$$

$$\hat{y} = 42.5818 - 0.6860 x \text{ (Estimated Regression Line)}$$

$$\text{Part (b)} \hat{y} = 42.5818 - 0.6860 (24.5) = 25.7748$$

11.5 Inferences Concerning the Regression Coefficients

Confidence Interval A $100(1 - \alpha)\%$ confidence interval for the parameter β_0 in the regression line for β_0 $\mu_{Y|x} = \beta_0 + \beta_1 x$ is

$$b_0 - t_{\alpha/2} \frac{s}{\sqrt{nS_{xx}}} \sqrt{\sum_{i=1}^n x_i^2} < \beta_0 < b_0 + t_{\alpha/2} \frac{s}{\sqrt{nS_{xx}}} \sqrt{\sum_{i=1}^n x_i^2},$$

where $t_{\alpha/2}$ is a value of the t -distribution with $n - 2$ degrees of freedom.

Confidence Interval A $100(1 - \alpha)\%$ confidence interval for the parameter β_1 in the regression line for β_1 $\mu_{Y|x} = \beta_0 + \beta_1 x$ is

$$b_1 - t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}} < \beta_1 < b_1 + t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}},$$

where $t_{\alpha/2}$ is a value of the t -distribution with $n - 2$ degrees of freedom.

11.17 With reference to Exercise 11.5 on page 398,

- (a) evaluate s^2 ;
- (b) construct a 95% confidence interval for β_0 ;
- (c) construct a 95% confidence interval for β_1 .

11.5 A study was made on the amount of converted sugar in a certain process at various temperatures. The data were coded and recorded as follows:

Temperature, x	Converted Sugar, y
1.0	8.1
1.1	7.8
1.2	8.5
1.3	9.8
1.4	9.5
1.5	8.9
1.6	8.6
1.7	10.2
1.8	9.3
1.9	9.2
2.0	10.5

$$b_0 = 6.41363 \text{ and } b_1 = 1.80909$$

$$\hat{y} = 6.41363 + 1.80909x$$

$$(a) s^2 = \frac{SSE}{n-2} = \frac{\sum(y-\hat{y})^2}{n-2} = \frac{3.60173}{9} = 0.40019,$$

$$s = 0.63261$$

$\hat{y}$							$(y - \hat{y})^2$
x	y	xy	x^2	y^2	ycap	(y-ycap)^2	
	1	8.1	8.1	1	65.61	8.22272	0.01506
	1.1	7.8	8.58	1.21	60.84	8.40363	0.36437
	1.2	8.5	10.2	1.44	72.25	8.58454	0.00715
	1.3	9.8	12.74	1.69	96.04	8.76545	1.07030
	1.4	9.5	13.3	1.96	90.25	8.94636	0.30652
	1.5	8.9	13.35	2.25	79.21	9.12727	0.05165
	1.6	8.6	13.76	2.56	73.96	9.30817	0.50151
	1.7	10.2	17.34	2.89	104.04	9.48908	0.50540
	1.8	9.3	16.74	3.24	86.49	9.66999	0.13689
	1.9	9.2	17.48	3.61	84.64	9.85090	0.42367
	2	10.5	21	4	110.25	10.03181	0.21920
Sum	16.5	100.4	152.59	25.85	923.58	100.39992	3.60173

For (b)

$$b_0 - t_{\alpha/2} \frac{s}{\sqrt{nS_{xx}}} \sqrt{\sum_{i=1}^n x_i^2} < \beta_0 < b_0 + t_{\alpha/2} \frac{s}{\sqrt{nS_{xx}}} \sqrt{\sum_{i=1}^n x_i^2},$$

where $t_{\alpha/2}$ is a value of the t -distribution with $n - 2$ degrees of freedom.

$$1 - \alpha = 0.95 \Rightarrow \alpha = 0.05, \frac{\alpha}{2} = 0.025$$

$$s = \sqrt{0.40079} = 0.63261$$

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} = 25.85 - \frac{16.5^2}{11} = 1.1$$

$$\text{Degree of freedom} = n - 2 = 9$$

$$t_{0.025,9} = 2.262$$

$$b_0 = 6.41363$$

$$6.41363 - 2.262 \frac{0.63261}{\sqrt{11 * 1.1}} \sqrt{25.85} < \beta_0 < 6.41363 + 2.262 \frac{0.63261}{\sqrt{11 * 1.1}} \sqrt{25.85}$$

$$4.324 < \beta_0 < 8.503$$

For (c)

Confidence Interval A $100(1 - \alpha)\%$ confidence interval for the parameter β_1 in the regression line
for β_1 $\mu_{Y|x} = \beta_0 + \beta_1 x$ is

$$b_1 - t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}} < \beta_1 < b_1 + t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}},$$

where $t_{\alpha/2}$ is a value of the t -distribution with $n - 2$ degrees of freedom.

$$1 - \alpha = 0.95 \Rightarrow \alpha = 0.05, \frac{\alpha}{2} = 0.025$$

$$s = \sqrt{0.40079} = 0.63261$$

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} = 25.85 - \frac{16.5^2}{11} = 1.1$$

$$\text{Degree of freedom} = n - 2 = 9$$

$$t_{0.025,9} = 2.262$$

$$b_1 = 1.80909$$

$$1.80909 - 2.262 \frac{0.63261}{\sqrt{1.1}} < \beta_1 < 1.80909 + 2.262 \frac{0.63261}{\sqrt{1.1}}$$
$$0.446 < \beta_1 < 3.172$$

Hypothesis Tests for the Slope of the Population Regression Line

$$T = \frac{(B_1 - \beta_1)/(\sigma/\sqrt{S_{xx}})}{S/\sigma} = \frac{B_1 - \beta_1}{S/\sqrt{S_{xx}}}$$

has a t -distribution with $n - 2$ degrees of freedom. The statistic T can be used to construct a $100(1 - \alpha)\%$ confidence interval for the coefficient β_1 .

Hypothesis Tests for the Intercept of the Population Regression Line

$$T = \frac{B_0 - \beta_0}{S \sqrt{\sum_{i=1}^n x_i^2 / (nS_{xx})}}$$

Important Formulas

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, \quad S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

OR

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}, \quad S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n}, \quad S_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n}$$

An unbiased estimate of σ^2 is

$$s^2 = \frac{SSE}{n-2} = \sum_{i=1}^n \frac{(y_i - \hat{y}_i)^2}{n-2} = \frac{S_{yy} - b_1 S_{xy}}{n-2}.$$

11.9 A study was made by a retail merchant to determine the relation between weekly advertising expenditures and sales.

Advertising Costs (\$)	Sales (\$)
40	385
20	400
25	395
20	365
30	475
50	440
40	490
20	420
50	560
40	525
25	480
50	510

- (b) Find the equation of the regression line to predict weekly sales from advertising expenditures.
- (c) Estimate the weekly sales when advertising costs are \$35.

$$(b) \sum_i x_i = 410, \sum_i y_i = 5445, \sum_i x_i^2 = 15,650, \sum_i x_i y_i = 191,325, n = 12. \text{ Therefore,}$$

$$b = \frac{(12)(191,325) - (410)(5445)}{(12)(15,650) - (410)^2} = 3.2208,$$

$$a = \frac{5445 - (3.2208)(410)}{12} = 343.7056.$$

$$\text{Hence } \hat{y} = 343.7056 + 3.2208x$$

$$(c) \text{ When } x = \$35, \hat{y} = 343.7056 + (3.2208)(35) = \$456.43.$$

11.21 Test the hypothesis that $\beta_1 = 6$ in Exercise 11.9 on page 399 against the alternative that $\beta_1 < 6$. Use a 0.025 level of significance.

$$s^2 = \frac{S_{yy} - b_1 S_{xy}}{n - 2}$$

11.21 The hypotheses are

$$H_0 : \beta = 6,$$

$$H_1 : \beta < 6.$$

$$T = \frac{(B_1 - \beta_1)/(\sigma/\sqrt{S_{xx}})}{S/\sigma} = \frac{B_1 - \beta_1}{S/\sqrt{S_{xx}}}$$

$$\alpha = 0.025.$$

Critical region: $t = -2.228$.

Computations: $S_{xx} = 15,650 - 410^2/12 = 1641.667$, $S_{yy} = 2,512.925 - 5445^2/12 = 42,256.25$, $S_{xy} = 191,325 - (410)(5445)/12 = 5,287.5$, $s^2 = \frac{42,256.25 - (3,221)(5,287.5)}{10} = 2,522.521$ and then $s = 50.225$. Now

$$t = \frac{3.221 - 6}{50.225/\sqrt{1641.667}} = -2.24.$$

Decision: Reject H_0 and claim $\beta < 6$.

11.8 A mathematics placement test is given to all entering freshmen at a small college. A student who receives a grade below 35 is denied admission to the regular mathematics course and placed in a remedial class. The placement test scores and the final grades for 20 students who took the regular course were recorded.

- (b) Find the equation of the regression line to predict course grades from placement test scores.

Placement Test	Course Grade
50	53
35	41
35	61
40	56
55	68
65	36
35	11
60	70
90	79
35	59
90	54
80	91
60	48
60	71
60	71
40	47
55	53
50	68
65	57
50	79

$$(b) \sum_i x_i = 1110, \sum_i y_i = 1173, \sum_i x_i^2 = 67,100, \sum_i x_i y_i = 67,690, n = 20. \text{ Therefore,}$$

$$b = \frac{(20)(67,690) - (1110)(1173)}{(20)(67,100) - (1110)^2} = 0.4711,$$

$$a = \frac{1173 - (0.4711)(1110)}{20} = 32.5059.$$

$$\text{Hence } \hat{y} = 32.5059 + 0.4711x$$

(d) If 60 is the minimum passing grade, below which placement test score should students in the future be denied admission to this course?

(d) For $\hat{y} = 60$, we solve $60 = 32.5059 + 0.4711x$ to obtain $x = 58.466$. Therefore, students scoring below 59 should be denied admission.

11.20 Test the hypothesis that $\beta_0 = 10$ in Exercise 11.8 on page 399 against the alternative that $\beta_0 < 10$. Use a 0.05 level of significance.

$$T = \frac{B_0 - \beta_0}{S \sqrt{\sum_{i=1}^n x_i^2 / (nS_{xx})}}$$

$$H_0 : \beta_0 = 10$$

$$H_a : \beta_0 < 10 \text{ (left-tailed)}$$

$$\alpha = 0.05$$

For a left-tailed test at $\alpha = 0.05$, $df = 18$:

$$t_{0.05,18} \approx -1.734$$

Computations: $S_{xx} = 67,100 - 1110^2/20 = 5495$, $S_{yy} = 74,725 - 1173^2/20 = 5928.55$,
 $S_{xy} = 67,690 - (1110)(1173)/20 = 2588.5$, $s^2 = \frac{5928.55 - (0.4711)(2588.5)}{18} = 261.617$ and then
 $s = 16.175$. Now

$$t = \frac{32.51 - 10}{16.175\sqrt{67,100/(20)(5495)}} = 1.78.$$

$$1.78 > -1.734$$

Fail to reject H_0

There is no evidence to support the claim that $\beta_0 < 10$.

Exercise questions (11.1 to 11.13) Page # 398
to 400

Exercise Questions: 11.17 – 11.21, page # 412